Software Requirements Specification

Student Digital Identity Wallet (T2-PS5)

1. INTRODUCTION

This document specifies the software requirements for the Student Digital Identity Wallet, a system designed to secure, store, and verify student academic credentials using cryptographic techniques. The goal is to eliminate document forgery and manual verification overhead by providing a tamper-evident digital wallet backed by public-key cryptography and QR-code-based public verification.

2. SCOPE

The system provides an end-to-end credential lifecycle management platform that covers:

- **Institution Portal:** Certificate generation/issuance, SHA-256 document hashing, digital signing, and automated QR code generation/stamping.
- **Student Wallet:** Secure repository where students view, download, and share their officially issued, QR-embedded digital certificates.
- **Verification Portal:** Publicly accessible endpoint for companies/verifiers to scan QR codes, validate cryptographic signatures, and re-hash document copies to detect physical or digital tampering.
- **Database & Security Layer:** SQLite persistence for user roles, public keys, cryptographic hashes, Base64 signatures, and issuance metadata.

2.1 OPERATING ENVIRONMENT

- **Programming Language:** Python 3.10+
- **Database Engine:** SQLite 3.x
- **Libraries:** hashlib (SHA-256 digests), cryptography (signing and verification), qrcode (QR rendering), Pillow (Image/PDF handling).
- **UI Framework:** Flask/FastAPI (Web-based portal) or CustomTkinter (Desktop GUI prototype)

3. CRYPTOGRAPHIC DATA INTEGRITY VERIFICATION:

- **Digital Signing (Asymmetric Encryption):** Signing data row hashes using a Private Key to guarantee authenticity and non-repudiation.
- **Signature Verification (PKI):** Using a Public Key (PEM format) to validate that data hasn't been modified since it was written.
- **Database Security & Integrity:** Protecting offline SQLite database records against local tampering or unauthorised modification.
- **Cryptographic Primitives:** SHA-256 Hashing, Base64 Encoding, and Public/Private Key Infrastructure (PKI).

4. SYSTEM FEATURES & FUNCTIONAL REQUIREMENTS

4.1 Admin Role (College / Issuing Authority)

Provides administrative functions for cryptographic key management, student account creation, certificate upload, digital signing, and automated QR code watermarking.

4.2 Functional Requirements

- The system shall allow logged-in Admin users to generate and register.
- The system shall allow Admins to enroll students and assign them unique Student IDs.
- The system shall allow Admins to issue a certificate by specifying Student ID, Certificate Title, and attaching the document file (.pdf, .png, .jpg).
- The system shall store the signed document, SHA-256 hash, Base64 signature string, and public key PEM in SQLite, pushing the final document to the target User's wallet.
- The system shall allow Admins to revoke certificates in cases of administrative correction or academic fraud.

4.3 User Role (Student)

- Provides students with a secure personal wallet to view, manage, and share their officially issued credentials.
- The system shall allow Users to authenticate with an assigned username/email and password.
- The system shall present a Digital Wallet displaying all certificates issued to the logged-in User's ID.
- The system shall display certificate metadata including Issuing Institution, Degree Title, Issue Date, and Verification Status.
- The system shall allow Users to copy direct verification links or export the standalone QR code image to share with recruiters.

4.4 Company Role (Verifier / Employer)

- Provides an unauthenticated public interface for external verifiers to validate document authenticity and perform live tamper tests.
- Upon receiving a verification request, the system shall fetch stored hash H, Base64 signature S, and Issuer Public Key from SQLite.
- The verification page shall provide a drag-and-drop file upload interface allowing companies to upload a candidate's shared file copy for live re-hashing.

5. CONCLUSION

The Student Digital Identity Wallet replaces slow, error-prone manual credential verification with instant, mathematically unforgeable cryptographic proof. Employers verify credentials in seconds by scanning an embedded QR code, eliminating the need to wait days or weeks for manual confirmation from an issuing college. Students maintain full ownership of a secure, shareable digital wallet containing all their official, tamper-evident academic records.